

Metal–Organic Dimerization of Dissymmetrical Ligands toward Customized Through-Space Chromophore Interactions

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Cite This: *Chem. Mater.* 2023, 35, 1788–1795



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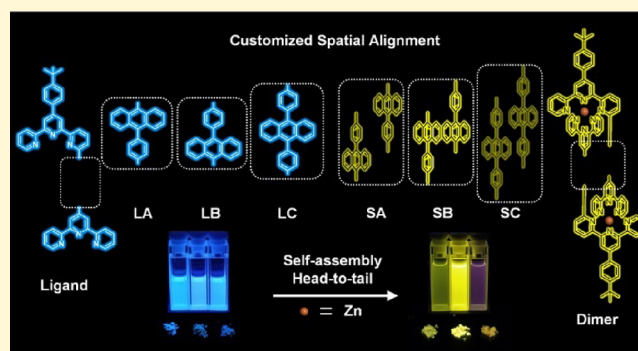


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Supporting Information

ABSTRACT: The pursue of good photophysical properties for organic optoelectronic materials requires a well understanding of through-space chromophore interactions, which further requires a well control over the spatial arrangement of chromophores. However, it remains a challenge to precisely customize the positioning of chromophores in their aggregating form such as in a simplest cofacially stacked dimer. Herein, this work provides a customizable molecular design based on dissymmetrical ligands that can enable a precise control over chromophore interactions through the formation of metal–organic dimers. Anti-paralleled stacking of two dissymmetrical ligands in the metal–organic dimers results in a lateral shifting of chromophore stacking, whose spacing is determined and adjusted by the degree of ligand dissymmetry. Three metal–organic dimers with a variation in chromophore spacing exhibited unique photophysical properties in both solution and solid states and displayed high-efficient luminescence against quenching in their aggregating states. This strategy thereby offers a universally applicable way to construct chromophore dimers with fixed cofacial spacing and determinate through-space interactions.



INTRODUCTION

Light-emitting materials have attracted considerable attention and been widely explored over the past few decades because of their diverse applications in light-emitting diodes,^{1–4} light-harvesting systems,^{5–7} biological sensing,^{8–12} imaging,^{13–15} etc. Interactions of chromophores directly have a significant effect on the photophysical properties of fluorescent materials.^{16–18} Kasha and co-workers provided the theoretical conception of molecular exciton theory in their pioneering work, suggesting that the photophysical properties in the solid state can be largely influenced by interactions between molecular transition dipoles.^{19–22} In some cases, crystals or solids with specific intermolecular interactions of chromophores exhibit tunable fluorescence properties.^{23–26} However, this strategy is diffusion-controlled and intermolecular interactions cannot be maintained when dispersed in solution, which limits the exploration of interaction–property relationships (Scheme 1a).

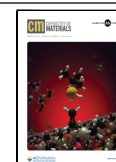
Dimers are probably the simplest and most practical system to enable the study of molecular interactions experimentally and theoretically, which often exhibit features not available in monomers.^{27,28} Various dimerization methods have been developed based on different driving forces including both non-covalent (hydrogen bonding,²⁹ macrocyclic complexations,^{30,31} host–guest interaction,³² halogen and chalcogen

bonding,³³ etc.) and covalent strategies.^{34,35} According to the level of dynamics and the nature of linkages, most dimers can be assigned to three categories: (i) dynamic noncovalent dimers, (ii) pseudostatic noncovalent dimers, and (iii) static covalent dimers.³⁶ These reported dimers have shown great applications in the fields of enhanced emission,^{37,38} room-temperature phosphorescence,^{39,40} supramolecular catalyst,^{41,42} and near-infrared probes.⁴³ However, there are still some great challenges remaining: (i) Modulation of the spatial alignments of chromophores within the dimer is not easy. Most of the dimerized chromophores performed as parallel and cofacial stacks with limited flexibility (Scheme 1a);^{44,45} (ii) in terms of the static covalent ones, the purification of targeting dimers against byproducts is often tedious because the covalent dimerization is not reversible and not self-adaptable.^{35,46,47} By contrast, pseudostatic noncovalent dimers can sustain a chromophore interaction for a longer period of time than the dynamic ones, while are still capable of self-correction on

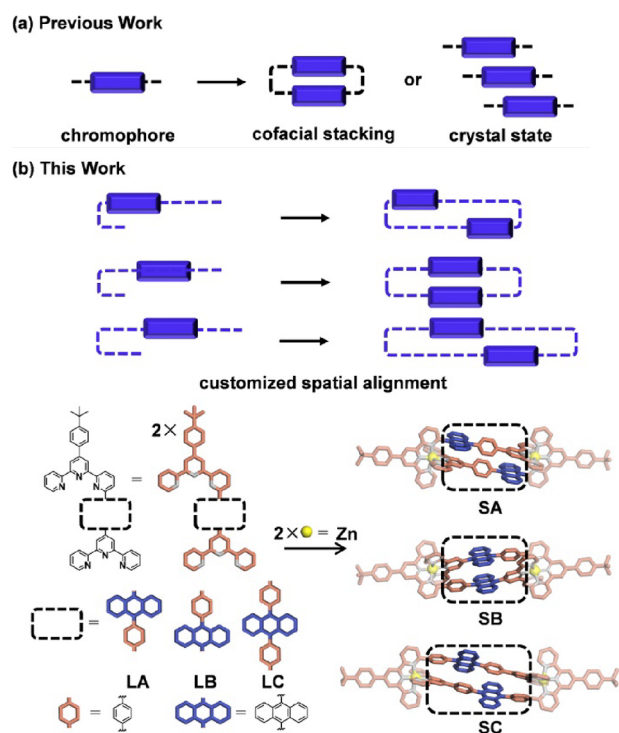
Received: December 8, 2022

Revised: January 31, 2023

Published: February 13, 2023



Scheme 1. (a) Common Strategies in the Literature Usually Fail to Modulate the Spatial Alignment of Chromophores. (b) Customized Strategy for Modulating Chromophore Interactions during Dimerization in the Current Work and Schematic Representation of the Formation of SA, SB, and SC



account of their noncovalent nature.⁴⁸ It is therefore important to develop customizable and simple strategies for the formation of stable pseudostatic dimers.

In this study, we have first provided a customizable strategy based on metal–ligand interactions for the construction of pseudostatic dimers capable of modulating chromophore spatial interactions (Scheme 1b). The coordination bond based on metal–ligand interactions has highly directional and predictable features, it has better stability than other non-covalent bonds, and it is widely used to construct 2D and 3D emissive metal–organic cages/cycles (MOCs) by introducing various luminescent motifs.^{49–53} In such reported MOCs, luminescent motifs could be accurately distributed on the final structures.^{54–57} However, the spatial interactions of the chromophores were neglected due to the customary symmetric design.⁴⁴ Herein, we have designed three dissymmetrical ditopic terpyridyl ligands LA, LB, and LC. This unique design features two terpyridine (tpy) units linked by different sequences of anthracene and benzene, leading to three different dimeric complexes SA, SB, and SC by head-to-tail coordination mode with Zn(II). Single-crystal X-ray diffraction data suggested that two anthracene units in a dimeric structure have entirely different intramolecular spatial alignments in three complexes, which led to distinct photoluminescence properties in the solution state. Complex SB with a maximum anthracene overlapping ratio exhibited the highest fluorescence efficiency. Moreover, all three complexes exhibited enhanced fluorescence quantum efficiency in the solid state compared with that in solution.

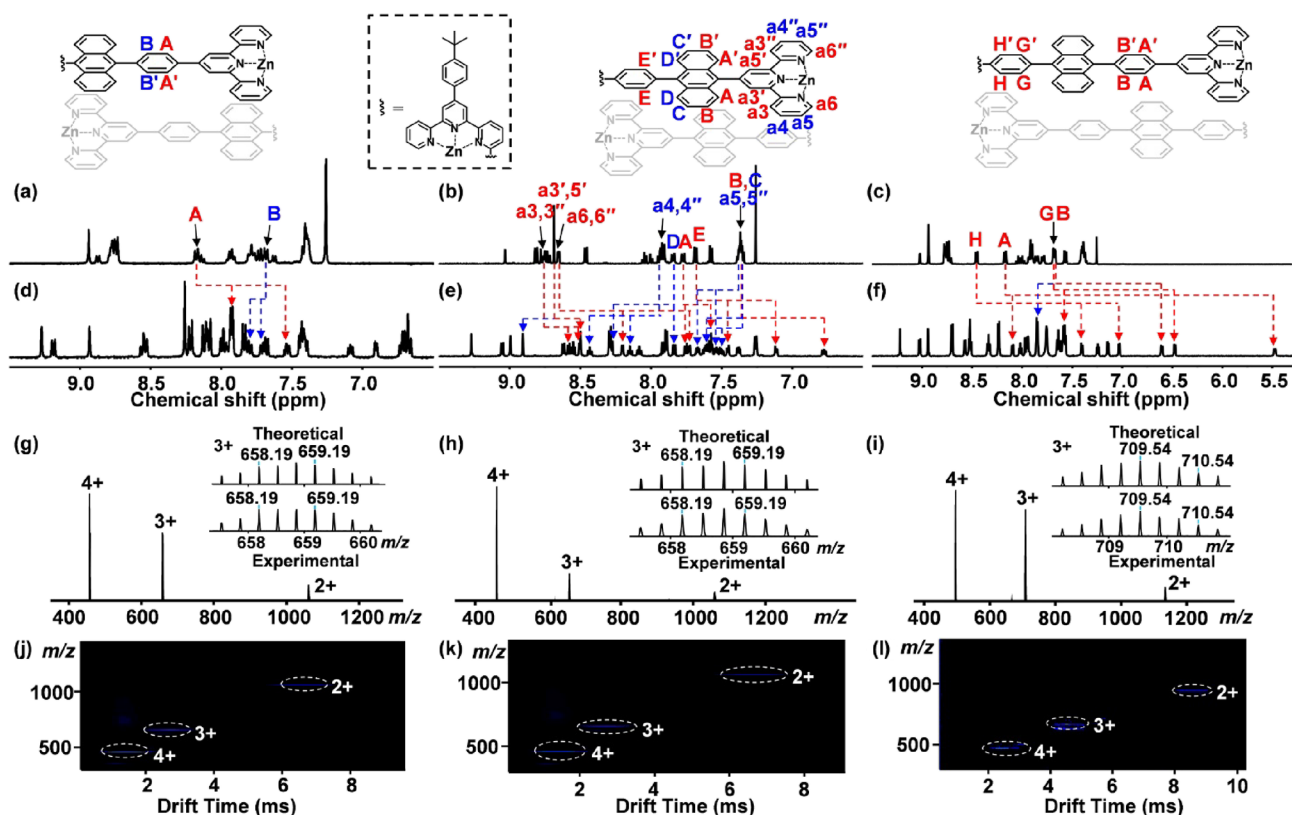


Figure 1. ¹H NMR spectra (500 MHz, 298 K) of (a) LA, (b) LB, and (c) LC in CDCl₃ and (d) SA, (e) SB, and (f) SC in CD₃CN. ESI-MS spectra (g–i) and TWIM-MS plots (*m/z* vs drift time) (j–l) of the assembly of SA, SB, and SC, respectively.

RESULTS AND DISCUSSION

Ligands **LA**, **LB**, and **LC** were designed by introducing different anthracene-based substituents between two tpy units and synthesized by Suzuki coupling reaction (Scheme S1). The key compound **3** with dissymmetrical modification was obtained by an efficient Kröhnke reaction of compounds **1** and **2**.⁵⁸ The yellow complexes were successfully prepared by mixing ligands and Zn(II) in an exact stoichiometric ratio of 1:1 in CHCl₃/CH₃OH ($v/v = 1:2$) solution at 50 °C for 8 h followed by addition of an excessive amount of NH₄PF₆ (yield >95%).

All compounds were characterized by ¹H, ¹³C, and 2D NMR (COSY and NOESY; see details in the Supporting Information) spectroscopy. As shown in Figure 1, ¹H NMR of all ligands shows five sets of signals for pyridine, which match the corresponding dissymmetrical structure. Interestingly, unlike ligands, the complexes exhibit a series of sharp and well-split signals with a 1:1 ratio attributed to the different chemical environments inside and outside the complexes caused by self-assembly.^{59,60} We speculate that the close arrangement of anthracene and benzene moieties restricted their rotation in each molecule.

Electrospray ionization mass spectrometry (ESI-MS) showed only one prominent set of peaks with different charge states for all three complexes resulting from the successive loss of different numbers of counterions (Figure 1g–i). After deconvolution, the obtained molecular weights of **SA**, **SB**, and **SC** were 2405.8, 2405.8, and 2558.0 Da, respectively, agreeing well with the expected chemical compositions of two ligands, two Zn(II) ions, and four PF₆[−]. Moreover, the experimental isotope patterns matched well with the simulated ones (Figures S66–S68). Traveling-wave ion mobility mass spectrometry (TWIM-MS) showed that each charge state of these dimers has a narrow drift time distribution (Figure 1j–l), indicating discrete and rigid assembly without other isomers or structural conformers.

To further confirm the molecular structure and configuration of the three complexes, single crystals of these dimers were obtained by slowly diffusing the vapor of the poor solvent into an acetonitrile solution of complexes for over 3 weeks. As expected, X-ray crystallographic analysis showed three different head-to-tail coordination modes (Figure 2 and Tables S1–S3). Noteworthy, the regular octahedral shape of (tpy-Zn(II)-tpy) deformed in all complexes (Figure S1), indicating that

substituents between two tpy units were restricted after self-assembly. Interplanar distances and overlaps proved as two important factors in evaluating the intramolecular interaction. For **SA**, a distance of 9.69 Å between two anthracene groups did not allow any overlapping. Each anthracene was positioned parallel to a pyridine moiety with an interlayer distance of 3.46 Å (Figure 2a,b). In contrast to **SA**, the two anthracenes were arranged face to face in **SB**, and their overlap ratio was 67%. For **SB**, the interplanar distance was 3.85 Å, indicating the existence of weak π – π interactions (Figure 2c,d). For **SC**, the distance on the short axis between two anthracenes was 5.14 Å. However, strong CH $\cdots\pi$ interactions between anthracene and adjacent benzene (2.55 Å; Figure S2) resulted in a huge dihedral angle and the longest interplanar distance between the two layers of anthracene (5.14 Å; Figure 2e,f). Notably, the crystallographic analysis showed huge dihedral angles in bridging moieties of all complexes (53.8° for **SA**, 73.9° for **SB**, and 78.0° for **SC**), which originated from the strong repulsion of hydrogen atoms between anthracene and adjacent phenyl. The observed different intramolecular interactions of anthracene allowed us to explore their photophysical properties.

After confirming the structure of all complexes, the photophysical properties of all ligands and complexes were investigated to clarify the structure–property relationship. The ligands showed almost identical UV–vis absorption and fluorescence spectra in CHCl₃ (Figure 3a). Their UV–vis absorption spectra clearly showed the features of tpy (250–340 nm) and typical vibrational absorption bands of anthracene moieties (358, 376, and 396 nm). The fluorescence spectra of all ligands revealed a strong emission centered at 437 nm in CHCl₃ (absolute fluorescence quantum yield $\Phi_F > 66\%$) originating from the ligand-localized π – π^* transitions in anthracene (electron donor) and tpy (electron acceptor) moieties due to the disappearance of the vibrational fine structures. These results implied that the luminescent properties of ligands are not affected by the positions of anthracene in a dilute solution. In contrast to the high fluorescence quantum yields in solution, powders exhibited typical aggregation-caused quenching behaviors in the solid state, as evidenced by a sharp decrease in their fluorescence quantum yields (Figure 3b,c) due to their rigid and conjugated molecular structures.

The optical properties of ligands motivated us to explore the optical properties of complexes. In a dilute solution (10.0 μ M), all three complexes displayed slightly red-shifted absorption bands and emission bands with two main parts (Figure 3d). The emission band at approximately 436 nm could be attributed to tpy. The emission at a longer wavelength originated due to the presence of a charge transfer excited state involving the anthracene moiety as an electron donor and the tpy moiety as an electron acceptor.⁶¹ Excitingly, the fluorescence spectra of the three complexes with different relative positions of anthracene fragments exhibited unexpected luminous properties. The **SA** revealed a moderate emission centered at 553 nm in acetonitrile with a Φ_F value of 6.7%, while **SB** exhibited a strong emission centered at 556 nm with a Φ_F value of 32.6%. The intramolecular relative position of anthracenes within these two isomeric complexes was the only factor affecting their fluorescence properties. The large overlap ratio of the two anthracenes restricts the intramolecular rotation or vibration of the rigid backbone, resulting in less energy loss via a non-radiative transition process. The fluorescence spectrum of **SC** showed a weak emission at 620

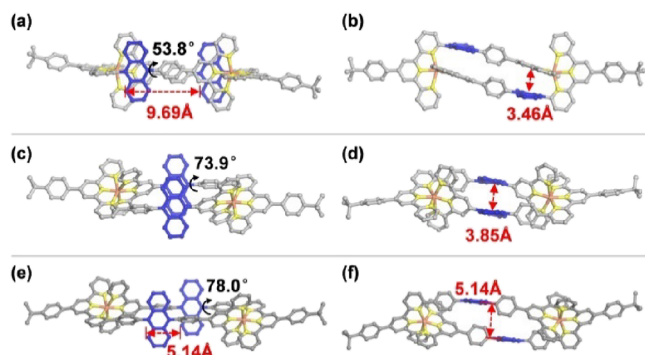


Figure 2. Top view (a, c, e) and front view (b, d, f) of the X-ray crystal structure of **SA**, **SB**, and **SC**, respectively. Hydrogen atoms, counterions, and solvents are omitted for clarity (C, gray or blue; N, yellow; Zn, orange).

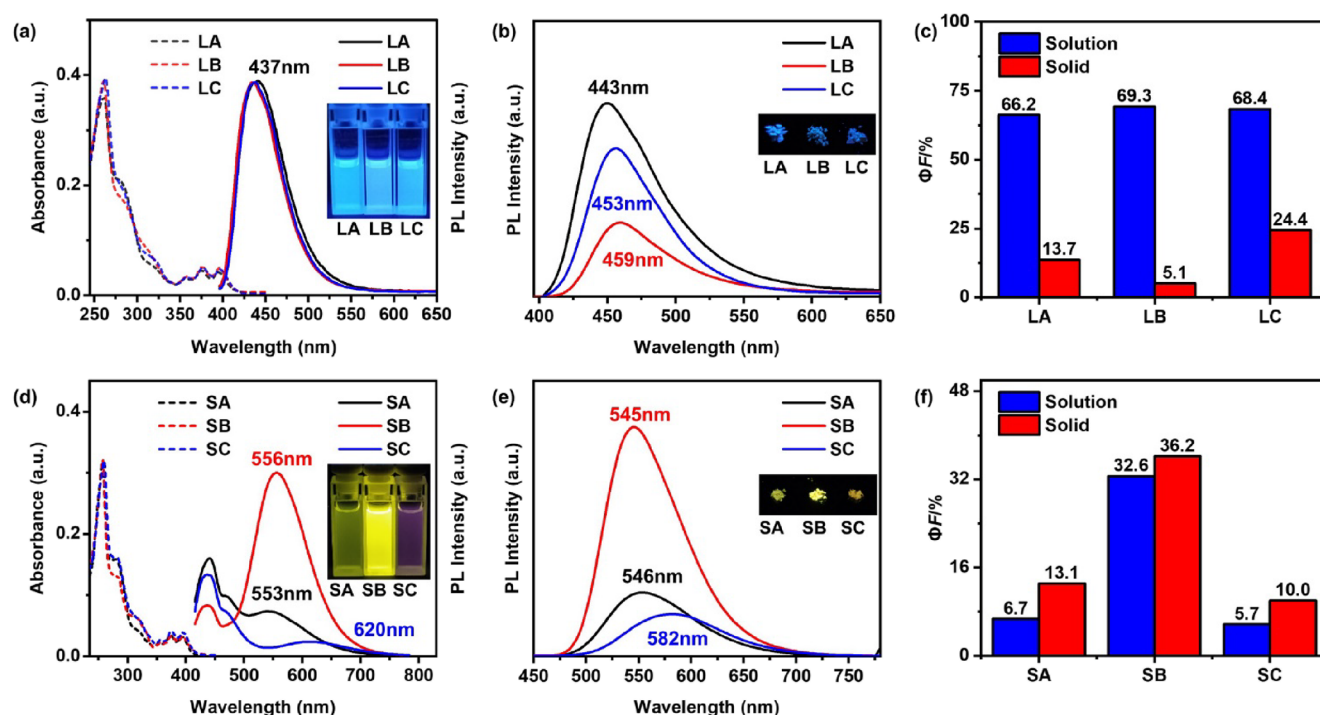


Figure 3. UV-vis absorption and fluorescence emission spectra of (a) three ligands in CHCl₃ and (b) three complexes in CH₃CN ($\lambda_{\text{ex}} = 396$ nm, $c = 10.0$ μM), and the insets show their solution photographs under 365 nm. Fluorescence spectra of (c) three ligands and (d) three complexes in the solid state ($\lambda_{\text{ex}} = 396$ nm), and their photographs under 365 nm are given in the insets. Φ_F of the three ligands (e) and complexes (f).

nm (Φ_F value of 5.7%), presumably because the additional benzene in SC effectively extended the molecular conjugation, resulting in largely red-shifted PL spectra. Moreover, the large distance between the two layers of anthracenes greatly reduced the steric hindrance. Consequently, although the anthracene groups in SC constituted a typical model of J-aggregation, favorable for the fluorescence emission, it exhibited almost no fluorescence.

Frontier molecular orbitals involved in the electronic transitions for the investigated molecules obtained by density functional theory (DFT) calculations are shown in Figure 4.

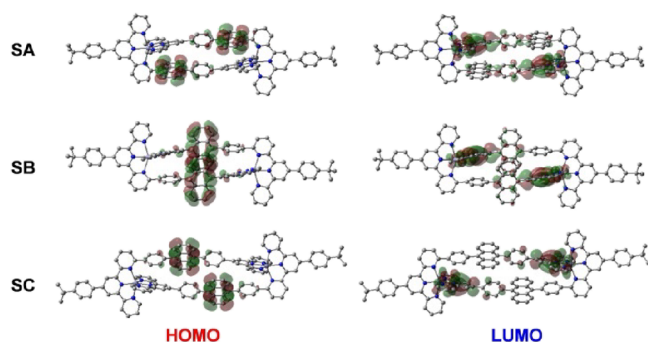


Figure 4. Frontier molecular orbitals involved in the electronic transitions for SA, SB, and SC obtained by DFT calculations.

The elections of SA, SB, and SC were mainly localized around their electron-donating anthracene in HOMO while located on top of the bridging moiety in LUMO, which suggests that dissymmetrical type at both ends does not participate in the charge transfer process. The calculated emission wavelengths (λ_{em}), excitation energies (E_{em}), oscillator strengths (f_{em}), and dominant orbital transitions are listed in Table S5. The

calculated emission wavelengths of SA, SB, and SC are 545, 525, and 620 nm, respectively, which are well consistent with the experimental data. Particularly, the emission oscillator strength (f_{em}) of SB is 0.2337, which is significantly larger than those of SA (0.0319) and SC (0.0018). The significantly larger f_{em} of SB can be ascribed to the relatively large HOMO-LUMO overlap, which is closely related to the position of anthracene. According to the Einstein spontaneous emission rate formula, the fluorescence radiative rate is positively correlated with the emission oscillator strength. This gives the qualitative explanation why SB has the relatively high fluorescence luminescence efficiency.

The fluorescence spectra and photographs of the complexes in the solid state are shown in Figure 3e. To our surprise, all three complexes exhibited higher fluorescence quantum yields in the solid state than in their dilute solutions (Figure 3f). Moreover, the emission maximum of SC showed a distinct blue shift compared with SA and SB during the transition from the solution state to the solid state. In solution, the difference in fluorescence properties of the three complexes originated from the intramolecular relative position of anthracene moieties, while in the solid state, the intermolecular relative position of the anthracene moieties was decisive for their fluorescence. As evident from Figure 5, SA, SB, and SC show one-dimensional long-range stacking. In SA and SB, anthracenes present on different molecules are far apart without overlap, implying that there is no interaction between them, and therefore, intramolecular interactions become responsible for the fluorescence properties of SA and SB. However, a shorter 2.65 Å interplanar distance between the two anthracenes of different molecules in SC leads to intermolecular interactions becoming a crucial factor for the fluorescence properties of SC. Complex SB exhibited strong emission behavior in both the solution and solid state, where

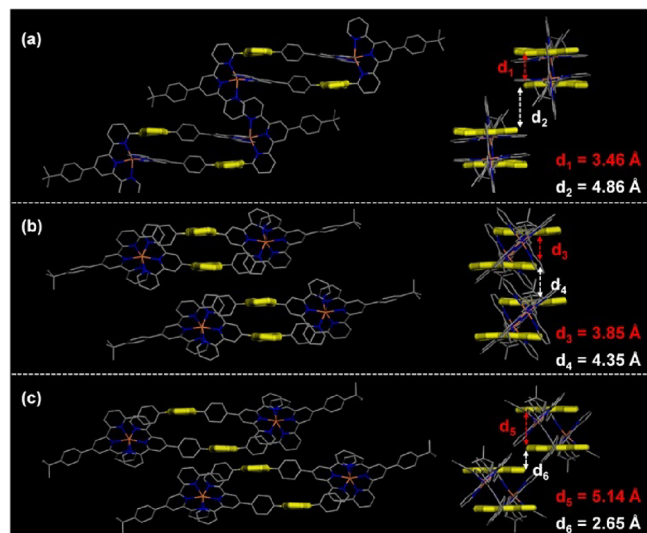


Figure 5. Schematic view of single-crystal stacking modes of (a) SA, (b) SB, and (c) SC.

weak intramolecular π – π interactions facilitated an enhanced emission induced by discrete dimer stacking, endowing SB with remarkable fluorescence properties.

CONCLUSIONS

In summary, we have reported a customizable strategy to realize precise control over the dimeric interactions of anthracenes as well as their fluorescence properties. The three dimers SA, SB, and SC were constructed via coordination-driven self-assembly using three dissymmetrical ligands LA, LB, and LC. From NMR spectroscopy, ESI-MS, and single-crystal analysis, the three complexes were identified as head-to-tail structures with the designed intramolecular relative position of anthracenes. The three dimers with different intramolecular interactions exhibited distinct luminescent properties in solution, and the unique complex SB with a maximum overlapping ratio exhibited the highest fluorescence efficiency. Particular intermolecular stacking modes of SA, SB, and SC resulted in enhanced fluorescence quantum efficiency in the solid state. More importantly, this is the first reported strategy based on metal–ligand interactions that can customize the spatial arrangement of anthracenes in a pseudostatic dimer. The customizable strategy provides a novel and universal platform for exploring the relationship between the interaction of chromophores and their photophysical behaviors and has extensive applicability for many types of chromophores.

EXPERIMENTAL SECTION

General Considerations. Reactions were performed under prepurified nitrogen using standard Schlenk techniques. NMR spectral data were recorded on Bruker 600 and 500 MHz and Qone AS 400-MHz nuclear magnetic resonance instruments. MALDI-TOF-MS was performed on a Bruker AutoflexIII using *trans*-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenyli-dene]malononitrile (DCTB) as a matrix. Both ESI-MS and TWIM-MS spectra were collected on a Waters Synapt G2 mass spectrometer. The UV–vis spectra were recorded on a Shimadzu UV2550 spectrophotometer for solution samples and a PerkinElmer Lambda 1050+ spectrophotometer for solid samples. Emission spectra and emission quantum yields were determined by a FLS920 spectrometer. X-ray diffraction

data were collected using synchrotron radiation and a MAR325 CCD detector at Shanghai Synchrotron Radiation BL17B Beamline.

Ligand LA. A mixture of compound 6 (0.50 g, 0.7 mmol, the synthetic route of compound 6 is shown in the Supporting Information), 4'-(4-boronatophenyl)[2,2':6',2'']terpyridine (0.30 g, 0.8 mmol), and $\text{Pd}(\text{PPh}_3)_2\text{Cl}_2$ (26.4 mg, 0.04 mmol) was degassed under nitrogen three times. After that, 40 mL of toluene, 20 mL of aqueous solution of 1 M Na_2CO_3 , and 10 mL of *tert*-butanol were added subsequently. The mixture was stirred under 75 °C for 24 h. After cooling down to room temperature, the mixture was extracted with CHCl_3 . The combined organic phase was washed with brine and dried over Na_2SO_4 . After removal of the solvent under vacuum, the residue was purified by silica gel chromatography with chloroform:methanol (100:1.2) as the eluent to afford a white solid (0.46 g, 75%), which was characterized by NMR and MALDI-TOF-MS.

Dimer SA. To a solution of ligand LA (6.1 mg, 7.18 μmol) in CHCl_3 (4 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (2.14 mg, 7.19 μmol) in MeOH (8 mL) was added; then, the mixture was stirred at 50 °C for 8 h. After cooling to room temperature, 200 mg of NH_4PF_6 was added and a yellow precipitate was observed. The precipitate was washed with water to afford the product (8.2 mg, 95%), which was characterized by NMR, ESI-MS, and TWIM-MS.

Ligand LB. The synthesis routes of precursor compound 4 and compound 8 are shown in the Supporting Information. A mixture of compound 4 (0.28 g, 0.49 mmol), compound 8 (0.26 g, 0.54 mmol), and $\text{Pd}(\text{PPh}_3)_2\text{Cl}_2$ (17.2 mg, 0.03 mmol) was degassed under nitrogen three times. After that, 60 mL of toluene, 30 mL of aqueous solution of 1 M Na_2CO_3 , and 15 mL of *tert*-butanol were added subsequently. The mixture was stirred under 75 °C for 24 h. After cooling down to room temperature, the mixture was extracted with CHCl_3 . The combined organic phase was washed with brine and dried over Na_2SO_4 . After removal of the solvent under vacuum, the residue was purified by silica gel chromatography with chloroform:methanol (100:1) as the eluent to afford a white solid (0.3 g, 71%), which was characterized by NMR and MALDI-TOF-MS.

Dimer SB. To a solution of ligand LB (9.0 mg, 10.60 μmol) in CHCl_3 (4 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (3.15 mg, 10.60 μmol) in MeOH (8 mL) was added; then, the mixture was stirred at 50 °C for 8 h. After cooling to room temperature, 200 mg of NH_4PF_6 was added and a yellow precipitate was observed. The precipitate was washed with water to afford the product (12.2 mg, 96%), which was characterized by NMR, ESI-MS, and TWIM-MS.

Ligand LC. The synthesis routes of precursor compound 4 and compound 9 are shown in the Supporting Information. A mixture of compound 4 (0.29 g, 0.52 mmol), compound 9 (0.32 g, 0.58 mmol), and $\text{Pd}(\text{PPh}_3)_2\text{Cl}_2$ (20.0 mg, 0.03 mmol) was degassed under nitrogen three times. After that, 40 mL of toluene, 20 mL of aqueous solution of 1 M Na_2CO_3 , and 10 mL of *tert*-butanol were added subsequently. The mixture was stirred under 75 °C for 24 h. After cooling down to room temperature, the mixture was extracted with CHCl_3 . The combined organic phase was washed with brine and dried over Na_2SO_4 . After removal of the solvent under vacuum, the residue was purified by silica gel chromatography with chloroform:methanol (100:1.5) as the eluent to afford a white solid (0.3 g, 62%), which was characterized by NMR and MALDI-TOF-MS.

Dimer SC. To a solution of ligand LC (9.7 mg, 10.48 μmol) in CHCl_3 (4 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (3.15 mg, 10.60 μmol) in MeOH (8 mL) was added; then, the mixture was stirred at 50 °C for 8 h. After cooling to room temperature, 200 mg of NH_4PF_6 was added and a yellow precipitate was observed. The precipitate was washed with water to afford the product (12.7 mg, 95%), which was characterized by NMR, ESI-MS, and TWIM-MS.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.chemmater.2c03668>.

Materials and methods, DFT calculations, and characterization data (PDF)

Crystallographic data of SA (CIF)

Crystallographic data of SB (CIF)

Crystallographic data of SC (CIF)

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Author Contributions

In this work, M.W. conceived and designed the experiments. K.L. completed the synthesis. K.L. and H.Y. grew the crystals and collected the X-ray data of the crystals. H.Y. and T.L. solved the X-ray structure of the crystals. H.Z. and J.L. performed the DFT calculations. K.L., H.Y., and N.H. conducted the photophysical property characterization. K.L., J.S., and N.H. conducted the NMR and MS characterization. M.W., G.W., H.Z., K.L., H.Y., J.S., and N.H. analyzed the data and wrote the manuscript. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We gratefully acknowledge the support from the National Natural Science Foundation of China (22271116 and 22071079 for M.W.) and the fellowship of China Postdoctoral Science Foundation (2021M701383 for J.S.) and thank the staff from BL17B Beamline of the National Facility for Protein Science in Shanghai (NFPS) at Shanghai Synchrotron Radiation Facility for assistance during data collection.

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